

Advanced Programming with Java – Exam Pattern Analysis

Trend window: last 3 sittings (Spring 2025, Fall 2024, Spring 2024)

Unit Weightage Table

Unit #	Unit Name	Total Appearances	Priority
1	Basics of Programming in Java	73	CRITICAL
2	Object Oriented Principles in Java	30	CRITICAL
3	Building Components using Swing and JavaFX	64	CRITICAL
4	Distributed Network Programming	36	CRITICAL
5	Database Connectivity with JAVA	39	CRITICAL
6	Servlets and JSP	9	MEDIUM
7	Advanced Topics in JAVA	13	HIGH

Must-Not-Miss Questions (High Probability Core)

Unit 2 – Object Oriented Principles in Java

- Explain Interfaces and Abstract Classes. How are Interfaces used to achieve Multiple Inheritance in Java? (Spring 2025, Fall 2024, Spring 2024, Fall 2023, Spring 2023, Fall 2021, Fall 2020, Fall 2019, Spring 2018, Fall 2018, Spring 2017, Spring 2016, Fall 2015, Spring 2015)(count: ~18) 🔥📈 ~
→ Examiner focus: Interface syntax, abstract methods, default methods, and resolving multiple inheritance ambiguity
- Explain Object-Oriented Inheritance, Runtime Polymorphism, and Method Overriding vs. Overloading in Java. (Spring 2025, Fall 2021, Spring 2018, Spring 2017, Fall 2016, Spring 2016, Spring 2023)(count: ~9) 🔥📈 ~
→ Examiner focus: Inheritance benefits, method signature rules, dynamic method dispatch, and code reusability

- Explain the implications of using the `final` keyword with variables, methods, and classes. (Spring 2025, Spring 2024, Spring 2020)(count: ~3) ★ 📈 ~
→ Examiner focus: Immutability, preventing method overriding, and restricting class inheritance

Unit 1 – Basics of Programming in Java

- Demonstrate File I/O operations and Stream handling (Byte/Character Streams, FileReader, FileWriter, Serialization) with proper exception handling. (Spring 2023, Fall 2022, Spring 2021, Fall 2021, Spring 2020, Fall 2020, Fall 2019, Spring 2018, Fall 2018, Spring 2017, Spring 2016)(count: ~11) 🔥 📈 ~
→ Examiner focus: File stream creation, reading bytes/characters until EOF, and Object Serialization syntax
- Explain Java Architecture (JVM, JRE, JDK), execution flow, and how Java achieves platform independence and architectural neutrality. (Spring 2025, Fall 2024, Spring 2024, Fall 2023, Spring 2023, Fall 2021, Fall 2020, Fall 2019, Spring 2018, Spring 2017, Spring 2015)(count: ~13) 🔥 📈 ~
→ Diagram compulsory
→ Examiner focus: Bytecode compilation, JVM execution engine, JIT compiler, and OS abstraction layer
- Explain Exception Handling in Java (try-catch-finally, throw/throws) and how to write custom exception classes. (Spring 2025, Spring 2024, Fall 2023, Fall 2022, Spring 2021, Fall 2021, Fall 2020, Spring 2018, Fall 2018, Spring 2017, Fall 2016, Spring 2016, Fall 2015, Spring 2015)(count: ~15) 🔥 📈 ~
→ Examiner focus: Exception hierarchy, checked vs unchecked exceptions, throw vs throws, and custom exception subclassing
- Explain the Reflection API in Java, its security implications, and how to inspect or access private fields and methods. (Spring 2023, Fall 2022, Spring 2021, Spring 2020, Fall 2020, Fall 2019, Fall 2018, Fall 2015, Spring 2015)(count: ~9) 🔥 📈 ~
→ Examiner focus: Introspection capabilities, accessing private constructors/methods, and Reflection security risks
- Explain Packages, ClassPath, and Access Modifiers (private, public, protected, default) in Java. (Spring 2025, Spring 2024, Fall 2023, Spring 2018, Spring 2016)(count: ~5) 🔥 📈 ~
→ Examiner focus: Package creation, package import syntax, member visibility scopes, and ClassPath variable configuration
- Explain constructors in Java, parameterized constructors, and types of constructors with examples. (Fall 2024, Fall 2018, Spring 2017)(count: ~3) ★ 📈 ~
→ Examiner focus: Object initialization, default vs parameterized constructors, and constructor overloading

Unit 4 – Distributed Network Programming

- Develop TCP Socket Client-Server applications in Java (e.g., string uppercase echo, square calculation, multi-user chat). (Spring 2025, Fall 2024, Spring 2024, Fall 2023, Fall 2022, Spring 2021, Fall 2021, Fall 2020, Fall 2019, Fall 2018, Spring 2017, Fall 2016, Fall 2015, Spring 2015)(count: ~16) 🔥 📊 ~
→ Diagram compulsory
→ Examiner focus: `ServerSocket` and `Socket` instantiation, network I/O streams, and connection lifecycle
- Explain Networking concepts in Java: IP Addresses (`InetAddress`), Ports, Socket types, and UDP Datagram programming. (Spring 2023, Fall 2022, Spring 2021, Spring 2018, Fall 2020, Fall 2015)(count: ~7) 🔥 📊 ~
→ Examiner focus: Connectionless communication, `DatagramSocket` , `DatagramPacket` , and host resolution
- Explain RMI Architecture, stubs and skeletons, and create RMI applications (e.g., factorial calculation, sum, odd/even check). (Spring 2025, Fall 2024, Spring 2024, Fall 2023)(count: ~5) 🔥 📊 ~
→ Examiner focus: Remote interfaces, `UnicastRemoteObject` , RMI registry lookup, and stub/skeleton communication
- Explain the URL and URLConnection classes with examples for reading web resource contents. (Fall 2020, Fall 2018, Spring 2016, Spring 2015)(count: ~4) 🔥 📊 ~
→ Examiner focus: Parsing URLs and opening stream connections to fetch HTTP resource contents
- Explain CORBA Architecture, IDL, and compare RMI vs CORBA. (Spring 2025, Spring 2024, Fall 2023)(count: ~3) ★ 📊 ~
→ Examiner focus: Language independence of CORBA vs Java-centric RMI architecture and IDL compilation

Unit 3 – Building Components using Swing and JavaFX

- Explain Applet Life Cycle, Sandbox Security model, converting Applets to Standalone Applications, and Applet GUI/Audio handling. (Spring 2023, Fall 2022, Spring 2021, Fall 2021, Fall 2020, Fall 2019, Spring 2018, Fall 2018, Spring 2017, Fall 2016, Spring 2016, Fall 2015, Spring 2015) (count: ~15) 🔥 📊 ~
→ Examiner focus: `init()` , `start()` , `stop()` , `destroy()` methods, HTML embedding, and sandbox security constraints

- Design interactive GUI Applications using Swing/JavaFX components (TextFields, Buttons, Labels, Frames) to process user input or perform calculations. (Spring 2025, Spring 2024, Fall 2023, Spring 2023, Fall 2020, Fall 2019, Spring 2018, Fall 2018, Spring 2017, Fall 2016, Spring 2016) (count: ~11) 🔥 📊 ~
→ Diagram compulsory
→ Examiner focus: Layout setup, component registration, event handling logic, and dynamic UI feedback
- Explain creation of Menus (JPopupMenu, JMenuBar), Dialog Boxes, and Closable Frames in Swing/AWT. (Fall 2022, Spring 2023, Fall 2018, Spring 2017, Fall 2016, Spring 2016, Fall 2015, Spring 2015)(count: ~10) 🔥 📊 ~
→ Examiner focus: Window closure event handling, menu hierarchies, and popup menu positioning
- Write Java Graphics programs to draw custom 2D shapes or data visualization charts (Pie Charts, Bar Charts). (Spring 2023, Fall 2022, Spring 2021, Fall 2021, Fall 2020, Fall 2018, Fall 2016, Spring 2015, Fall 2015)(count: ~9) 🔥 📊 ~
→ Diagram compulsory
→ Examiner focus: Custom component drawing, `paintComponent()` , `Graphics2D` methods (`drawRect` , `fillArc`), and chart scaling
- Explain Swing Layout Management (FlowLayout, GridLayout, GridBagLayout, BoxLayout) with example programs. (Spring 2025, Fall 2024, Spring 2023, Spring 2021, Fall 2021, Spring 2020, Fall 2020, Fall 2019, Spring 2015)(count: ~9) 🔥 📊 ~
→ Diagram compulsory
→ Examiner focus: Container layout mechanics, grid coordinate constraints, and responsive positioning
- Explain Swing and AWT Event Handling mechanism, Event Delegation Model, and Listener Interfaces. (Fall 2024, Spring 2023, Fall 2021, Spring 2017, Spring 2016, Fall 2019, Fall 2015)(count: ~7) 🔥 📊 ~
→ Examiner focus: Event source, event listener registration, and event object handling routines
- Compare JavaFX vs Swing and explain JavaFX Stage, Scene, and JavaFX Layout Managers. (Fall 2024, Spring 2024, Fall 2023)(count: ~3) ★ 📊 ~
→ Examiner focus: Scene Graph architecture, Stage/Scene organization, and modern UI controls vs Swing

Unit 5 – Database Connectivity with JAVA

- Explain JDBC Architecture, JDBC Driver Types (Type 1 to Type 4), and the steps to connect a Java application to a database. (Fall 2024, Spring 2024, Fall 2022, Spring 2021, Fall 2021, Spring 2020, Fall 2020, Fall 2019, Spring 2018, Fall 2018, Fall 2016, Spring 2016, Fall 2015, Spring 2023)(count:

~15) 🔥 📈 ~

→ Diagram compulsory

→ Examiner focus: JDBC Driver categorization (Pure Java vs Native bridge),

`DriverManager.getConnection()` , and connection workflow steps

- Write Java Programs to perform database CRUD operations (SELECT, INSERT, UPDATE, DELETE) using JDBC Statements. (Spring 2025, Fall 2023, Fall 2022, Spring 2021, Fall 2021, Spring 2020, Fall 2020, Fall 2019, Spring 2018, Fall 2018, Spring 2017, Fall 2015, Spring 2015, Spring 2023)(count:

~14) 🔥 📈 ~

→ Examiner focus: Executing queries (`executeQuery` , `executeUpdate`), handling

`ResultSet` , and database resource cleanup

- Differentiate between `Statement` and `PreparedStatement` . Explain SQL Injection attacks and how PreparedStatements prevent them. (Spring 2025, Fall 2024, Spring 2024, Fall 2023, Spring 2018, Spring 2015)(count: ~6) 🔥 📈 ~

→ Examiner focus: Pre-compiled SQL performance advantages, parameter binding (? placeholders), and escaping malicious input

→ 🔗 Often paired with: Explain SQL Injection attack and how PreparedStatements prevent it.

- Explain `ResultSet`, `ResultSetMetaData`, `RowSets`, and SQL Escapes in JDBC. (Spring 2025, Fall 2016, Spring 2017, Spring 2016)(count: ~4) 🔥 📈 ~

→ Examiner focus: Column metadata extraction, dynamic table schema inspection, and SQL escape sequences

Unit 7 – Advanced Topics in JAVA

- Explain Object-Relational Mapping (ORM) and the Hibernate framework. Compare ORM with standard JDBC. (Spring 2025, Fall 2024, Spring 2024, Fall 2023)(count: ~5) 🔥 📈 ~

→ Examiner focus: Mapping relational schemas to Java classes, session management, and productivity advantages over JDBC

- Explain Design Patterns in Java, focusing on Singleton Pattern and Factory Method Pattern. (Spring 2025, Fall 2024, Spring 2024, Fall 2023)(count: ~4) 🔥 📈 ~

→ Examiner focus: Creational pattern implementations, private constructors, static instance getters, and factory object instantiation

→ 🔗 Often paired with: Compare or implement Singleton and Factory Method Design Patterns.

- Define Multithreading in Java. Explain thread creation methods (anonymous classes, Thread subclass, Runnable interface) and multithreaded synchronization. (Spring 2025, Fall 2024, Fall 2022)(count: ~3) ★ 📈 ~

→ Examiner focus: Concurrent thread execution, thread lifecycle states, `start()` vs `run()` , and anonymous class thread instantiation

Unit 6 – Servlets and JSP

- Explain Servlet Life Cycle (`init` , `service` , `destroy`), HTTP methods (`doGet` / `doPost`), and writing web servlets. (Fall 2024, Spring 2024, Fall 2023, Spring 2025)(count: ~5) 🔥 📈 ~
→ Examiner focus: Servlet container invocation cycle, request-response handling, and dynamic server-side processing
- Explain Cookie and Session Management in Java Servlets with code examples. (Spring 2025, Spring 2024, Fall 2023)(count: ~3) ★ 📈 ~
→ Examiner focus: Client-side persistent cookies vs server-side `HttpSession` objects, cookie creation, and session tracking

Unit 1: Basics of Programming in Java

Unit Overview: Exam questions heavily test Java execution mechanics, exception handling mechanisms, and File I/O stream operations, frequently asking for sample code alongside architectural definitions.

- Explain Exception Handling in Java (try-catch-finally, throw/throws) and how to write custom exception classes. (count: ~15) 🔥 📊 ~
 - "Define exception and explain why it is necessary to handle exceptions? WAP to write bytes of data to a file and reading it until end of file is obtained, using proper exception handling." (Fall 2022, Spring 2021, Fall 2020)(count: 3) ★
 - "How do you create own exception subclasses? Explain with an example." (Fall 2021, Spring 2018, Spring 2015)(count: 3) ★
 - "Define Exception handling with Example." (Spring 2017, Spring 2016)(count: 2) ★
 - "Write down the code to handle user defined exception for LowBalance when balance is less than 10 rupees on your mobile." (Spring 2025)(count: 1)
 - "Write a program with custom exception handler that handle arithmetic exception." (Spring 2024)(count: 1)
 - "Differentiate between error and exception." (Fall 2023)(count: 1)
 - "Write a program to handle ArithmeticException." (Fall 2018)(count: 1)
 - "Differentiate between checked and unchecked exceptions with relevant examples." (Fall 2016)(count: 1)
 - "What are exceptions? Write a program to read a string from the console and display it in the console." (Fall 2015)(count: 1)
 - "Write short notes on Custom Exception." (Fall 2022)(count: 1)
→ Examiner focus: Exception hierarchy, try-catch-finally blocks, throw vs throws, and subclassing `Exception`
- Explain Java Architecture (JVM, JRE, JDK), execution flow, and how Java achieves platform independence and architectural neutrality. (count: ~13) 🔥 📊 ~
 - "Define the term Architectural Neutral / Explain how Java is platform independent." (Spring 2025, Fall 2021, Spring 2018, Spring 2017)(count: 4) 🔥
 - "Explain the architecture of Java, including the role of JVM, JRE, and JDK." (Spring 2024, Fall 2023, Fall 2020, Spring 2015)(count: 4) 🔥
 - "What is the role of the JVM in Java Architecture? Describe the steps to compile and run a Java program." (Fall 2024)(count: 1)
 - "Explain the innovation and principles of java programming language that makes it runnable on any platform, with a suitable diagram." (Spring 2023, Fall 2019)(count: 2) ★

- "Write short notes on: How Java becomes Platform Independent? / JVM / JRE." (Spring 2021, Spring 2017, Spring 2016)(count: 2) ★
 - Diagram compulsory
 - Examiner focus: Role of JVM bytecode execution engine, JIT compiler, and hardware abstraction layer
- Demonstrate File I/O operations and Stream handling (Byte/Character Streams, FileReader, FileWriter, Serialization) with proper exception handling. (count: ~11) 🔥 📊 ~
 - "Write a complete Java program to write primitive data to file through Serialization mechanism." (Spring 2023)(count: 1)
 - "What are the types of I/O streams? Write a program in java using FileWriter to write string in a new file." (Spring 2020)(count: 1)
 - "Create a program to write string inside a file abc.txt." (Fall 2019, Spring 2016)(count: 2) ★
 - "Explain about FileReader and BufferedWriter class." (Fall 2021, Spring 2018)(count: 2) ★
 - "A data file emp.txt contains name, address and salary. Write a program to display specific records." (Fall 2018)(count: 1)
 - "Write a program to read content from file abc.txt and store it in xyz.txt." (Spring 2017)(count: 1)
 - Examiner focus: Byte vs Character stream classes, FileReader/FileWriter syntax, and `Serializable` marker interface
- Explain the Reflection API in Java, its security implications, and how to inspect or access private fields and methods. (count: ~9) 🔥 📊 ~
 - "Does the use of Reflection API challenges the security of Java? Explain with proper example." (Spring 2023)(count: 1)
 - "What are the pros and cons of Reflection API? Write a sample program that does reflective operation of accessing a private method." (Fall 2022)(count: 1)
 - "Write a program in Java to access a constructor, a private field and a private method using Reflection API." (Spring 2021)(count: 1)
 - "Why Reflection is regarded as a powerful aspect/tool in software testing and debugging? Elaborate with an example." (Spring 2020, Fall 2020)(count: 2) ★
 - "What is reflection? Why do we need reflection? Explain with sample code." (Spring 2015)(count: 1)
 - "Write short notes on Reflection API." (Fall 2019, Fall 2018, Fall 2015)(count: 3) ★
 - Examiner focus: Runtime introspection using `Class.forName()` , `getDeclaredMethod()` , and `setAccessible(true)`
- Explain Packages, ClassPath, and Access Modifiers (private, public, protected, default) in Java. (count: ~5) 🔥 📊 ~
 - "Define ClassPath? Explain its significance in Java." (Spring 2024, Fall 2023)(count: 2) ★

- "What is package in java? How can you create user-defined package in java? Explain with example." (Spring 2025, Fall 2023, Spring 2016)(count: 3) ★
- "Explain the various access modifiers in Java and their scope (private, public, protected, default)." (Spring 2024, Fall 2023, Spring 2018)(count: 3) ★
→ Examiner focus: Package syntax, import resolution, ClassPath environment configuration, and member access boundaries
- Explain constructors in Java, parameterized constructors, and types of constructors with examples. (count: ~3) ★ 📊 ~
 - "What is a constructor in Java? Explain the different types of constructors with examples." (Fall 2024)(count: 1)
 - "What is constructor, Explain with example?" (Spring 2017)(count: 1)
 - "Create parameterized constructor to initialize instance variables and override toString()." (Fall 2018)(count: 1)
→ Examiner focus: Default vs parameterized constructors, constructor chaining, and object state initialization
- Basic Java Syntax, Wrapper Classes, Arrays, and main() method signature. (count: ~16) 🔥 📊 ~
 - "Explain public static void main(String[] args)." (Fall 2022, Spring 2021)(count: 2) ★
 - "Write a Java program to perform array operations (delete element / separate even and odd)." (Fall 2022, Spring 2021)(count: 2) ★
 - "What are wrapper classes? Explain." (Fall 2021, Fall 2020, Spring 2018)(count: 3) ★
 - "Write a program to show usages of various data types in java." (Spring 2017)(count: 1)
 - "Define java. Write a program to print Hello Java in console." (Spring 2016)(count: 1)
 - "What is a class file? Differentiate between assignment and initialization." (Fall 2015, Fall 2016)(count: 2) ★
 - "Illustrate the scope of static and local variable with an example." (Spring 2020)(count: 1)
 - "Write short notes on: Inner Class / History of Java / Benefits of Java / C++ vs Java / JIT compiler." (Fall 2021, Fall 2018, Spring 2018, Fall 2015, Spring 2015)(count: 4) 🔥
→ Examiner focus: Entry point method signature components, primitive object wrapping, and basic array indexing
- Explain the Java Collections Framework and its core components (List, Set, Map). (Spring 2024) (count: 1)
→ Examiner focus: Differences between List, Set, and Map interfaces and implementation classes

Unit Summary: total unique questions: 8; top frequent: Exception Handling (15), Java Architecture (13), File I/O Operations (~11)

Unit 2: Object Oriented Principles in Java

Unit Overview: Interface vs Abstract class and Multiple Inheritance constitute the absolute core of this unit, appearing in almost every single sitting.

- Explain Interfaces and Abstract Classes. How are Interfaces used to achieve Multiple Inheritance in Java? (count: ~18) 🔥 📈 ~
 - "Define Interface. As we know, there is no room for multiple inheritance using classes. How do you achieve multiple inheritance in java. Explain with Example." (Spring 2025, Fall 2024, Fall 2023, Fall 2020, Fall 2019, Fall 2018, Spring 2015)(count: 7) 🔥
 - "Differentiate between abstract class and interface in Java with examples." (Spring 2025, Fall 2023, Fall 2021, Spring 2018, Fall 2015)(count: 5) 🔥
 - "Define abstraction / interface. Write Java program that override methods getUserValue() and displayUserDetail()." (Spring 2024, Fall 2023)(count: 2) ★
 - "What is an Interface? Explain why interfaces are necessary with suitable programming example." (Spring 2023, Spring 2016)(count: 2) ★
 - "What is difference between Interface and Class? Explain with an example." (Spring 2017)(count: 1)
 - "Write short notes on Abstract Class." (Fall 2021)(count: 1)
→ Examiner focus: Interface contract definition, `implements` vs `extends`, abstract method declaration, and multiple interface implementation
- Explain Object-Oriented Inheritance, Runtime Polymorphism, and Method Overriding vs. Overloading in Java. (count: ~9) 🔥 📈 ~
 - "Explain the concept of inheritance in Java. What are the benefits of using inheritance, and how does it promote code reusability and polymorphism? Provide an example." (Spring 2025)(count: 1)
 - "Define subclass and super class. Explain inheritance hierarchy." (Fall 2023, Spring 2017, Fall 2016)(count: 3) ★
 - "What is method overloading? Can you override a private or static method in Java? Explain with an example." (Fall 2021, Spring 2018)(count: 2) ★
 - "Define Polymorphism. How should we implement polymorphism in java?" (Spring 2016)(count: 1)
 - "Define Inheritance and Composition. Illustrate the difference between them." (Fall 2021)(count: 1)
 - "Write short notes on Runtime Polymorphism." (Spring 2023, Fall 2016)(count: 1)
→ Examiner focus: Dynamic method dispatch, `@Override` semantics, overloading signature rules, and static/private binding constraints

- Explain the implications of using the `final` keyword with variables, methods, and classes.

(count: ~3) ★  ~

- "Explain final classes and methods." (Spring 2025)(count: 1)
- "What are the implications in Java when defining a variable, method, and class as final?" (Spring 2024)(count: 1)
- "The final keyword in Java can be applied to fields, parameters, methods and classes. Explain what the effect of final keyword is." (Spring 2020)(count: 1)
→ Examiner focus: Constant values, preventing method overriding, and forbidding class inheritance

Unit Summary: total unique questions: 3; top frequent: Interface & Multiple Inheritance (18), Inheritance & Polymorphism (9), Final Keyword (~3)

Unit 3: Building Components using Swing and JavaFX

Unit Overview: Questions heavily alternate between traditional Swing components/Applets and modern JavaFX UI design, with a strong focus on layout managers and event-driven GUI programs.

- Explain Applet Life Cycle, Sandbox Security model, converting Applets to Standalone Applications, and Applet GUI/Audio handling. (count: ~15) 🔥 📊 ~
 - "Define Applet and explain its life cycle in detail." (Spring 2023, Fall 2020, Fall 2019, Spring 2017, Fall 2016, Spring 2016)(count: 6) 🔥
 - "What are the differences between applet and normal java program? Can we convert an application to applet / applet to application?" (Spring 2023, Spring 2018, Fall 2018, Fall 2015)(count: 4) 🔥
 - "Create an applet that plays audio with play, stop, repeat GUI functionality." (Fall 2021, Spring 2018)(count: 2) ★
 - "Explain Sandbox Security model / display picture in an applet." (Spring 2015)(count: 2) ★
 - "Write short notes on Life cycle of an applet / Applet Skeleton." (Fall 2021, Fall 2020)(count: 1)
 - Examiner focus: Applet initialization lifecycle callbacks (`init` , `start` , `stop` , `destroy`), HTML applet tag, and browser security constraints
- Design interactive GUI Applications using Swing/JavaFX components (TextFields, Buttons, Labels, Frames) to process user input or perform calculations. (count: ~11) 🔥 📊 ~
 - "Write JavaFX code to design a login page testing username PU and password PU@2025Spring." (Spring 2025)(count: 1)
 - "Create GUI application with buttons RED and GREEN changing each other's background color." (Spring 2024)(count: 1)
 - "Write a program to create a Frame with two TextFields, Label, and Button to calculate sum of two values." (Fall 2023, Spring 2017, Fall 2018)(count: 3) ★
 - "Frame with two TextFields checking if mouse pointer is inside/outside and showing coordinates." (Fall 2020)(count: 1)
 - "Swing GUI with button and text fields checking even or odd number." (Fall 2019)(count: 1)
 - "Swing GUI converting textfield content to uppercase on button click." (Spring 2018, Spring 2016)(count: 2) ★
 - "Frame with two buttons BLACK and BLUE changing background color." (Fall 2016)(count: 1)
 - "Swing Frame with 3 text fields and menu add, subtract, close." (Fall 2015)(count: 1)
 - Diagram compulsory
 - Examiner focus: Container component hierarchy, `ActionListener` attachment, input validation, and visual UI layout

- Creation of Menus (JPopupMenu, JMenuBar), Dialog Boxes, and Closable Frames in Swing/AWT. (count: ~10) 🔥 📊 ~
 - "How to close AWT/Swing Window in Java? Explain methods and write code for closable frame." (Spring 2023, Fall 2022, Spring 2015)(count: 3) ★
 - "Create a popup menu attached to frame displaying selection message on label." (Fall 2022) (count: 1)
 - "What is a dialog box? Explain its types and program to create own dialog box." (Fall 2018) (count: 1)
 - "Differentiate between Frame and Swing / Panel vs Frame / AWT vs Swing." (Spring 2017, Fall 2016, Spring 2015)(count: 3) ★
 - "Create a frame with title, height 400, width 400." (Spring 2016)(count: 1)
 - "Write short notes on Closable Frames." (Spring 2023, Fall 2016)(count: 1)
 - Examiner focus: `WindowListener` , `setDefaultCloseOperation()` , `JMenuBar` / `JPopupMenu` construction, and heavyweight vs lightweight components
- Explain Swing Layout Management (FlowLayout, GridLayout, GridBagLayout, BoxLayout) with example programs. (count: ~9) 🔥 📊 ~
 - "Why do we need layout management? Explain GridBag layout with suitable example." (Spring 2025)(count: 1)
 - "Compare different layout managers in Java Swing with examples." (Fall 2024)(count: 1)
 - "State layout managers in swing. WAP to generate output using BoxLayout." (Spring 2023, Spring 2020, Fall 2020)(count: 3) ★
 - "Differentiate between Grid Layout and GridBag Layout / FlowLayout." (Spring 2021, Fall 2019, Spring 2015)(count: 3) ★
 - "Frame with two TextFields and Button arranged in GridLayout." (Spring 2023)(count: 1)
 - Diagram compulsory
 - Examiner focus: Layout manager choice, `GridBagConstraints` positioning rules, and Box layout axis alignments
- Write Java Graphics programs to draw custom 2D shapes or data visualization charts (Pie Charts, Bar Charts). (count: ~9) 🔥 📊 ~
 - "Write Java program using AWT/Swing to draw Pie Chart representing data/survey results." (Spring 2021, Fall 2021, Fall 2020)(count: 3) ★
 - "Write Java program using Graphics to draw Bar Chart representing statistics." (Spring 2021, Fall 2018, Spring 2015)(count: 3) ★
 - "Draw a 2D Square in red, Circle in green inside, and text above / Circular object moving with directional buttons." (Spring 2023, Fall 2022)(count: 2) ★
 - "Draw 2D rectangle in green with Flag of Nepal inside and text MY NEPAL." (Fall 2016)(count: 1)
 - Diagram compulsory

→ Examiner focus: `paint(Graphics g)` overriding, arc angle calculations for pie charts, and `drawRect` / `fillRect` dimensions

- Explain Swing and AWT Event Handling mechanism, Event Delegation Model, and Listener Interfaces. (count: ~7) 🔥 📊 ~
 - "Explain event handling mechanism in Java. Write a program using ActionListener." (Fall 2024, Spring 2016)(count: 2) ★
 - "What do you mean by delegation event model? Explain event listener interfaces." (Fall 2021, Fall 2015)(count: 2) ★
 - "Repaint method relation with update() and paint(). Example of repaint()." (Fall 2015)(count: 1)
 - "Write short notes on Event Delegation Model / Events and Event Classes / Event Handling." (Spring 2023, Fall 2019, Spring 2017)(count: 2) ★
→ Examiner focus: Event Source, Event Object, Event Listener interface contracts, and callback event processing
- Compare JavaFX vs Swing and explain JavaFX Stage, Scene, and JavaFX Layout Managers. (count: ~3) ★ 📊 ~
 - "Define stage and scene in JavaFX. Differentiate between Swing and JavaFX." (Fall 2024) (count: 1)
 - "Pros and Cons of JavaFX. Explain any two JavaFX layout managers." (Spring 2024, Fall 2023)(count: 2) ★
→ Examiner focus: JavaFX hardware acceleration, FXML decoupling, Scene graph hierarchy, and Stage window container

Unit Summary: total unique questions: 7; top frequent: Applet Lifecycle (15), Interactive GUI Applications (11), Menus & Dialogs (~10)

Unit 4: Distributed Network Programming

Unit Overview: TCP socket client-server programming dominates the long questions, while RMI applications and CORBA comparisons consistently appear across recent paper sittings.

- Develop TCP Socket Client-Server applications in Java (e.g., string uppercase echo, square calculation, multi-user chat). (count: ~16) 🔥 📊 ~
 - "Create a TCP client server application where client sends a string and server responds by echoing string in upper case." (Spring 2025, Fall 2023)(count: 2) ★
 - "Write TCP client-server application using TCP sockets, explain flow of communication." (Fall 2024, Spring 2021, Fall 2016, Fall 2015)(count: 4) 🔥
 - "TCP client server program where client sends integer value and server responds with squared value." (Spring 2024, Fall 2019)(count: 2) ★
 - "Create TCP socket program to check if client string is Palindrome or not on server." (Fall 2021)(count: 1)
 - "Create a TCP client/server program in which multiple users can chat / multithreaded TCP server." (Fall 2022, Spring 2021)(count: 2) ★
 - "UNIX client-server interaction diagram with detailed methods." (Fall 2019, Fall 2020)(count: 2) ★
 - "TCP client sending message / string input to server." (Fall 2018, Spring 2017, Spring 2015)(count: 3) ★
 - Diagram compulsory
 - Examiner focus: `ServerSocket.accept()` , `Socket` stream binding, `BufferedReader/PrintWriter` data transfer, and socket exception handling
- Explain Networking concepts in Java: IP Addresses (`InetAddress`), Ports, Socket types, and UDP Datagram programming. (count: ~7) 🔥 📊 ~
 - "Explain about sockets. Write a program to display IP address of a computer / Netflix IP list." (Spring 2023, Fall 2022)(count: 2) ★
 - "Explain UDP protocol. Write steps to create client server program using UDP / Datagram classes." (Spring 2023, Spring 2018)(count: 2) ★
 - "Differentiate between TCP and UDP sockets / Datagram vs Stream." (Fall 2022, Spring 2018, Fall 2018, Fall 2016)(count: 2) ★
 - "Write short notes on TCP and UDP sockets / Connection-oriented vs connection-less." (Spring 2021, Fall 2020)(count: 1)
 - Examiner focus: `InetAddress.getByName()` , `DatagramSocket` , `DatagramPacket` buffer setup, and connectionless protocol mechanics

- Explain RMI Architecture, stubs and skeletons, and create RMI applications (e.g., factorial calculation, sum, odd/even check). (count: ~5) 🔥 📊 ~
 - "What is the significance of stub and skeleton in RMI? Create RMI app to calculate factorial of given integer." (Spring 2025)(count: 1)
 - "Develop RMI application where client input two numbers if both numbers are even then result $(a+b)^2$ otherwise $(a+b)^3$." (Fall 2024)(count: 1)
 - "Explain RMI architecture. Develop RMI app where client request to identify given number is Odd or Even." (Spring 2024)(count: 1)
 - "Define RMI stubs and skeleton. Create RMI app where client remotely invokes method sending sum of two integers." (Fall 2023)(count: 1)
 - Examiner focus: Remote Interface contract extending `Remote` , `RemoteException` , `Naming.rebind()` , and stub-skeleton proxy mechanism
- Explain the URL and URLConnection classes with examples for reading web resource contents. (count: ~4) 🔥 📊 ~
 - "Explain the use of URL and URL Connection class with a suitable example/program." (Fall 2020, Fall 2018, Spring 2015)(count: 3) ★
 - "Write short notes on URL." (Spring 2016)(count: 1)
 - Examiner focus: `URL` object parsing, `openConnection()` , `URLConnection.getInputStream()` , and reading HTTP header/body streams
- Explain CORBA Architecture, IDL, and compare RMI vs CORBA. (count: ~3) ★ 📊 ~
 - "Write short notes on CORBA." (Spring 2025, Fall 2023)(count: 2) ★
 - "Write short notes on RMI vs. CORBA." (Spring 2024)(count: 1)
 - Examiner focus: CORBA ORB architecture, OMG IDL language mapping, and multi-language interoperability vs RMI Java-only model
- Explain JavaMail API and protocols used in JavaMail API. (Fall 2023)(count: 1)
 - Examiner focus: SMTP, POP3, IMAP protocols, `Session` , `Message` , and `Transport` class usages

Unit Summary: total unique questions: 6; top frequent: TCP Socket Applications (16), Networking & UDP (7), RMI Applications (~5)

Unit 5: Database Connectivity with JAVA

Unit Overview: Database driver architecture, PreparedStatement to mitigate SQL Injection, and hands-on DDL/DML CRUD coding questions form the backbone of JDBC exam questions.

- Explain JDBC Architecture, JDBC Driver Types (Type 1 to Type 4), and the steps to connect a Java application to a database. (count: ~15) 🔥 📊 ~
 - "What are the main components of JDBC architecture? Explain role of DriverManager and Connection interface." (Fall 2024)(count: 1)
 - "What is a JDBC driver? Explain different JDBC driver types (Type 1 to Type 4) and configuration." (Spring 2024, Fall 2022, Spring 2021, Spring 2020, Fall 2020, Fall 2019, Spring 2018, Fall 2016, Fall 2015)(count: 9) 🔥
 - "Write all steps to connect Java Application with any Database with sample program." (Spring 2024, Spring 2020, Fall 2021, Spring 2023)(count: 4) 🔥
 - "Differences between JDBC and ODBC / JDBC-ODBC deprecation." (Fall 2021, Spring 2023)(count: 1)
 - "Write short notes on JDBC API / Types of JDBC drivers." (Fall 2019, Spring 2018, Fall 2018)(count: 2) ★
 - Diagram compulsory
 - Examiner focus: `Class.forName()` , `DriverManager.getConnection()` , JDBC Driver layers, and pure Java vs bridge drivers
- Write Java Programs to perform database CRUD operations (SELECT, INSERT, UPDATE, DELETE) using JDBC Statements. (count: ~14) 🔥 📊 ~
 - "A table student consists of id, name and program. Write program asking program name and displaying total enrolled students / all student records." (Spring 2025, Fall 2023)(count: 2) ★
 - "Console application with CREATE, READ, UPDATE and DELETE queries using PreparedStatement." (Spring 2024)(count: 1)
 - "Insert student detail into database table student_tbl / Prompt y/n after entry." (Spring 2024, Fall 2021, Fall 2020, Spring 2023)(count: 4) 🔥
 - "Database table query: total count and average salary / average mark." (Fall 2022, Spring 2021, Fall 2019)(count: 3) ★
 - "Write program to update salary to 50000 where post is Manager / update student table." (Spring 2018, Spring 2017)(count: 2) ★
 - "Display all records from database table employee." (Fall 2018, Fall 2015, Spring 2015)(count: 2) ★
 - Examiner focus: Executing `executeQuery()` for SELECT, `executeUpdate()` for DML statements, iterating `ResultSet` , and closing resources

- Differentiate between `Statement` and `PreparedStatement` . Explain SQL Injection attacks and how PreparedStatements prevent them. (count: ~6) 🔥 📈 ~
 - "What is SQL Injection Attack? Write sample program using prepared statement to prevent SQL Injection Attack." (Spring 2025, Fall 2023)(count: 2) ★
 - "Differentiate between Statement and PreparedStatement. Demonstrate use with examples." (Fall 2024, Spring 2024, Spring 2018, Spring 2015)(count: 4) 🔥
 - 🔗 Often paired with: Explain SQL Injection attack and how PreparedStatements prevent it.
 - Examiner focus: Precompiled SQL queries, escaping parameterized values via ? placeholders, and query execution efficiency
- Explain ResultSet, ResultSetMetaData, RowSets, and SQL Escapes in JDBC. (count: ~4) 🔥 📈 ~
 - "What is ResultSetMetaData? Provide simple example code to illustrate its usage." (Fall 2016) (count: 1)
 - "Write short notes on SQL Escapes." (Spring 2025)(count: 1)
 - "Write short notes on ResultSet." (Spring 2017, Spring 2016)(count: 2) ★
 - Examiner focus: Dynamic database table schema discovery via `ResultSetMetaData` and database-neutral escape syntax

Unit Summary: total unique questions: 4; top frequent: JDBC Architecture & Drivers (15), ~~CRUD Database Programs~~ (14), Statement vs PreparedStatement & SQL Injection (~6)

Unit 6: Servlets and JSP

Unit Overview: Exam topics focus on web application lifecycle fundamentals, HTTP GET vs POST handling, and managing client session state using Cookies and HttpSession.

- Explain Servlet Life Cycle (`init` , `service` , `destroy`), HTTP methods (`doGet` / `doPost`), and writing web servlets. (count: ~5) 🔥 📊 ~
 - "What are differences between static and dynamic web apps? How does servlet fit into dynamic web application?" (Fall 2024)(count: 1)
 - "Discuss life cycle of a servlet in detail including methods `init()`, `service()`, and `destroy()`." (Fall 2024, Spring 2024, Fall 2023)(count: 3) ★
 - "Explain use of `doGet()` and `doPost()` methods of Servlet with an example." (Spring 2024, Fall 2023)(count: 2) ★
 - "Servlet program accepting input string from form and checking if palindrome." (Fall 2023) (count: 1)
 - "Write short notes on Handling GET/POST Requests." (Fall 2024)(count: 1)
→ Examiner focus: Servlet container invocation callbacks, HTTP request protocol translation, and `HttpServletRequest` / `HttpServletResponse` lifecycle
- Explain Cookie and Session Management in Java Servlets with code examples. (count: ~3) ★ 📊 ~
 - "What is cookie? How can you store and read cookies by using servlet. Explain with proper example." (Spring 2025)(count: 1)
 - "Create Servlet application to showcase concept of session and cookies." (Spring 2024, Fall 2023)(count: 2) ★
→ Examiner focus: Client-side `Cookie` header manipulation vs server-side `HttpSession.setAttribute()` , session timeout, and persistence
- Differentiate between Servlets and JSP. Form processing using JSP and Servlets. (count: ~2) ★ 📊 ~
 - "Differentiate between servlet and JSP. Create HTML form (First/Last name) and JSP file to read and display full name." (Spring 2025)(count: 1)
 - "Write short notes on Advantages of MVC." (Fall 2024)(count: 1)
→ Examiner focus: Presentation logic separation in JSP vs controller handling in Servlets, form parameter parsing, and MVC design model

Unit Summary: total unique questions: 3; top frequent: Servlet Lifecycle & Methods (5), ~~Cookie & Session Management~~ (3), JSP vs Servlet (~2)

Unit 7: Advanced Topics in JAVA

Unit Overview: Recently added advanced frameworks, design patterns, ORM abstraction, and multithreaded concurrency appear as high-value choices in modern paper sittings.

- Explain Object-Relational Mapping (ORM) and the Hibernate framework. Compare ORM with standard JDBC. (count: ~5) 🔥 📊 ~
 - "Write down the advantage of ORM over JDBC." (Spring 2025)(count: 1)
 - "Write short notes on Hibernate / Hibernate ORM." (Spring 2025, Fall 2024)(count: 2) ★
 - "Write short notes on Overview of ORM." (Spring 2024, Fall 2023)(count: 2) ★
 - Examiner focus: Relational table to Java class mapping, boilerplate JDBC reduction, dialect handling, and SessionFactory management
- Explain Design Patterns in Java, focusing on Singleton Pattern and Factory Method Pattern. (count: ~4) 🔥 📊 ~
 - "Explain Singleton Pattern and Factory method pattern in detail / Compare Singleton with Factory pattern." (Spring 2025, Fall 2024, Fall 2023)(count: 3) ★
 - "What are design patterns? Why do we need design patterns?" (Fall 2024, Fall 2023)(count: 2) ★
 - "Write short notes on Design patterns." (Spring 2024)(count: 1)
 - 🔗 Often paired with: Compare or implement Singleton and Factory Method Design Patterns.
 - Examiner focus: Creational design patterns, private constructor control, lazy static initialization, and abstract factory creation interfaces
- Define Multithreading in Java. Explain thread creation methods (anonymous classes, Thread subclass, Runnable interface) and multithreaded synchronization. (count: ~3) ★ 📊 ~
 - "Define multithreading. Explain techniques to create thread using anonymous class." (Spring 2025)(count: 1)
 - "Write multithreaded program in Java to print even and odd numbers alternatively." (Fall 2024)(count: 1)
 - "Differentiate between process and thread. How multithreading is performed in java? Illustrate with example." (Fall 2022)(count: 1)
 - Examiner focus: Concurrent execution model, extending `Thread` vs implementing `Runnable` , inline anonymous inner class syntax, and thread synchronization

Unit Summary: total unique questions: 3; top frequent: ORM & Hibernate (5), Design Patterns (4), Multithreading (~3)

Short-Note Questions (Repeating, Low-Frequency)

Count 1

- Java Collections Framework components (List, Set, Map). (Spring 2024)(count: 1)
- JavaMail API and email protocols (SMTP, POP3, IMAP). (Fall 2023)(count: 1)
- Write short notes on Advantages of MVC. (Fall 2024)(count: 1)
- Write short notes on SQL Escapes. (Spring 2025)(count: 1)
- Write short notes on Custom Exception. (Fall 2022)(count: 1)
- Write short notes on Abstract Class. (Fall 2021)(count: 1)
- Write short notes on RMI vs. CORBA. (Spring 2024)(count: 1)
- Write short notes on Overview of ORM. (Spring 2024)(count: 1)
- Scope of static and local variables in Java. (Spring 2020)(count: 1)
- Types of constructors in Java with examples. (Fall 2024)(count: 1)
- Form processing using JSP and HTML files. (Spring 2025)(count: 1)
- Write short notes on C++ vs Java. (Spring 2018)(count: 1)
- Write short notes on History of Java. (Fall 2018)(count: 1)
- Write short notes on Benefits of Java. (Fall 2015)(count: 1)
- Write short notes on JIT compiler. (Spring 2015)(count: 1)
- Write short notes on JRE. (Spring 2017)(count: 1)

Count 2

- JavaFX Stage and Scene definition and JavaFX Layouts. (Fall 2024, Spring 2024)(count: 2) ★
- Package creation and importing user-defined packages. (Spring 2025, Fall 2023)(count: 2) ★
- ClassPath definition and configuration in Java. (Spring 2024, Fall 2023)(count: 2) ★
- Method overloading and rules for overriding private/static methods. (Fall 2021, Spring 2018)(count: 2) ★
- UDP protocol mechanics and socket steps. (Spring 2023, Spring 2018)(count: 2) ★
- InetAddress class and retrieving network IP addresses. (Spring 2023, Fall 2022)(count: 2) ★
- Servlet doGet() and doPost() method execution. (Spring 2024, Fall 2023)(count: 2) ★
- Write short notes on CORBA. (Spring 2025, Fall 2023)(count: 2) ★
- Write short notes on Hibernate / Hibernate ORM. (Spring 2025, Fall 2024)(count: 2) ★
- Write short notes on Closable Frames. (Spring 2023, Fall 2016)(count: 2) ★
- Write short notes on ResultSet. (Spring 2017, Spring 2016)(count: 2) ★

Count 3

- Implications of using the final keyword with fields, methods, and classes. (Spring 2025, Spring 2024, Spring 2020)(count: 3) ★ 
- Parameterized constructors and toString() method overriding. (Fall 2024, Fall 2018, Spring 2017) (count: 3) ★ 
- Cookie and Session Management in Servlets. (Spring 2025, Spring 2024, Fall 2023)(count: 3) ★ 
- Multithreading concepts and thread creation (anonymous class / Runnable / Thread). (Spring 2025, Fall 2024, Fall 2022)(count: 3) ★ 
- Access modifiers scope and visibility (private, public, protected, default). (Spring 2024, Fall 2023, Spring 2018)(count: 3) ★
- RMI application implementation (factorial, even/odd, sum). (Spring 2025, Fall 2024, Spring 2024) (count: 3) ★ 
- JDBC PreparedStatement vs Statement and SQL Injection prevention. (Spring 2025, Fall 2024, Spring 2024)(count: 3) ★ 
- Write short notes on Design Patterns (Singleton and Factory method). (Spring 2025, Fall 2024, Spring 2024)(count: 3) ★ 

Cram Sheet

- Interfaces and Abstract Classes: Differences, definitions, and achieving Multiple Inheritance in Java. 🔥📈
- TCP Socket Client-Server Applications: String uppercase echo, square calculation, and multi-user chat. 🔥📈
- Applet Life Cycle, Sandbox Security model, Standalone conversion, and embedding Audio/Graphics. 🔥📈
- JDBC Architecture, Driver Types (Type 1 to Type 4), and Java Database Connection Workflow. 🔥📈
- Exception Handling in Java: try-catch-finally, throw/throws, checked vs unchecked, and custom exception classes. 🔥📈
- Database CRUD Operations: Executing SELECT, INSERT, UPDATE, DELETE queries using JDBC Statements. 🔥📈
- Java Architecture: JVM, JRE, JDK, execution pipeline, platform independence, and architectural neutrality. 🔥📈
- Interactive Swing & JavaFX GUI Applications: Input processing, calculations, and dynamic UI updates. 🔥📈
- File I/O Operations: Byte/Character Streams, FileReader, FileWriter, and Object Serialization mechanics. 🔥📈
- Menus, Dialog Boxes, Popup Menus, and Closable Frame implementation in Swing/AWT. 🔥📈
- Object-Oriented Principles: Inheritance hierarchies, Runtime Polymorphism, and Overriding vs Overloading. 🔥📈
- Reflection API: Inspecting classes, modifying private fields/methods, and Reflection security risks. 🔥📈
- Swing Layout Management: FlowLayout, GridLayout, GridBagLayout, and BoxLayout containers. 🔥📈
- Graphics Programming in Java: Custom 2D shape drawing, Pie Charts, and Bar Chart statistics rendering. 🔥📈
- Networking Basics: IP resolution (`InetAddress`), Socket types, and UDP Datagram programming. 🔥📈
- Event Handling Mechanisms: Delegation Event Model, ActionListeners, and Event Listener Interfaces. 🔥📈
- JDBC PreparedStatement vs Statement: Pre-compiled queries and preventing SQL Injection attacks. 🔥📈
- Servlet Life Cycle (`init` , `service` , `destroy`), HTTP methods (`doGet` / `doPost`), and dynamic Web Applications. 🔥📈
- Packages, ClassPath environment variable setup, and Access Modifiers scope. 🔥📈
- Object-Relational Mapping (ORM) and Hibernate Framework vs Standard JDBC. 🔥📈

- URL and URLConnection Classes: Reading and parsing contents of web URL endpoints. 🔥 
- Design Patterns: Creational pattern principles, Singleton Pattern, and Factory Method Pattern. 🔥 
- Implications of using the `final` keyword with variables, methods, and classes. ★ 
- Parameterized Constructors, default constructors, and constructor overloading rules. ★ 
- CORBA Architecture, OMG IDL language, and RMI vs CORBA architecture comparison. ★ 
- JavaFX Architecture: Stage, Scene hierarchy, FXML integration, and JavaFX layout managers. ★ 
- Cookie and Session Management in Servlets: Client cookies vs server-side `HttpSession`. ★ 
- Multithreading in Java: Thread lifecycle, anonymous class thread creation, and concurrent execution. ★ 